

# ZAINAB IRFAN

AI/ML Engineer • RAG & LLM Systems Specialist • Full-Stack Python Developer

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## PROFESSIONAL PROFILE

Results-driven AI/ML Engineer with hands-on, production-grade experience designing and shipping Retrieval-Augmented Generation (RAG) systems, conversational LLM agents, and full-stack AI platforms. Proven track record delivering end-to-end solutions — from data pipelines and model integration to deployed, customer-facing applications — using FastAPI, Next.js, LangChain, and vector databases such as Qdrant. Combines strong software engineering fundamentals with applied machine learning expertise, backed by 7 industry certifications from AWS, IBM, and DeepLearning.AI. Adept at translating business requirements into scalable, reliable AI products and comfortable owning projects independently in fast-paced, client-facing environments.

## CORE TECHNICAL EXPERTISE

- **AI / LLM Engineering** — RAG Pipeline Architecture · LangChain · Prompt Engineering · LLM Fine-Tuning (LoRA/QLoRA, PEFT) · LLM Evaluation & Benchmarking · Conversational AI Agents · Streaming Chat Interfaces · Workflow Automation (n8n)
- **Machine Learning** — Supervised & Unsupervised Learning · Feature Engineering · Model Evaluation · Bias Detection · Exploratory Data Analysis (EDA)
- **Languages & Frameworks** — Python · JavaScript · HTML/CSS · FastAPI · Streamlit · Next.js · React · PyTorch · Transformers
- **Data & Vector Infrastructure** — Qdrant · Supabase · NumPy · Pandas · Scikit-learn · Matplotlib · Seaborn
- **APIs, Cloud & Deployment** — REST API Design & Integration · AWS (AI/ML) · Render Deployment · Infrastructure Automation · Google Colab (NVIDIA T4 GPU)
- **Languages Spoken** — English (Proficient) · Urdu (Native)

## WORK EXPERIENCE

### Freelance AI/ML Engineer & Full-Stack Developer

2024 – Present

Self-Employed / Independent Contractor

- Designed and built multiple production-style RAG platforms and LLM-powered applications, including document intelligence systems, conversational support agents, and workflow automation tools, using FastAPI, Next.js, LangChain, and Qdrant/Supabase.
- Owned the full software development lifecycle independently — API design, data pipeline development, model integration, cloud deployment (Render/AWS), and post-launch iteration — while managing client requirements and delivery timelines.
- Applied machine learning and data analysis techniques (EDA, feature engineering, model evaluation) to deliver predictive insights for clients across e-commerce and clinical/healthcare data use cases.
- Built automation systems (e.g., LLM-driven email response pipelines with n8n) that reduced manual operational workload for client businesses.

## KEY PROJECTS

### Prism AI — RAG-Based Document Intelligence Platform

2026

**Stack:** Next.js, FastAPI, LangChain, Qdrant, Supabase, OpenAI/Claude APIs · Deployed on Render

- Built a full-stack Retrieval-Augmented Generation platform that ingests multi-format documents (PDF, DOCX, PPT, CSV) and returns structured, context-aware answers.
- Designed the retrieval pipeline with LangChain and Qdrant for semantic search, using Supabase for persistent storage of documents and chat history.
- Integrated LLM APIs (OpenAI and Claude) with streamed chat responses for a real-time conversational Q&A experience.
- Built an analytics dashboard visualizing usage insights, queries per day, question volume, and engagement trends, with interactive charts.
- Deployed a production build with a Next.js frontend and a FastAPI backend hosted on Render.

**Skills Applied:** RAG Pipeline Design, LangChain, Vector Search (Qdrant), FastAPI, Next.js, LLM API Integration, Streaming Interfaces, Cloud Deployment, Data Visualization

### Efficient LLM Fine-Tuning using LoRA & QLoRA (TinyLlama-1.1B-Chat)

2026

**Stack:** Python, PyTorch, Transformers, PEFT, TRL, BitsAndBytes, Hugging Face Datasets · Google Colab (NVIDIA T4 GPU)

- Fine-tuned **TinyLlama-1.1B-Chat** for **Text-to-SQL** generation using **PEFT** techniques, including **LoRA** and **QLoRA**, on ~3,000 curated examples from the Hugging Face b-mc2/sql-create-context dataset.
- Trained on **Google Colab** with an **NVIDIA T4 GPU** across 3 epochs, reaching **84.2% validation accuracy**.
- Applied **4-bit quantization** and **low-rank adaptation** to substantially cut **GPU memory usage** and training cost while preserving model performance.

**Skills Applied:** LLM Fine-Tuning, LoRA, QLoRA, PEFT, 4-bit Quantization, GPU Optimization, Hugging Face, PyTorch, Text-to-SQL

### AI Customer Support Agent for Shopify Stores

2025 – 2026

**Stack:** Python, LLM APIs, Shopify Integration

- Developed an AI-powered customer support agent that integrates with a Shopify storefront to handle customer queries automatically.
- Used LLM APIs to generate context-aware responses, reducing manual reply workload for store operations.

**Skills Applied:** LLM API Integration, Conversational AI Agents, Third-Party API Integration, Python

### Automated Email Reply System

2025

**Stack:** Python, LLM APIs, Email Automation, n8n

- Built an automation system that reads incoming emails, generates context-appropriate draft replies using an LLM, and sends responses automatically.

**Skills Applied:** Workflow Automation (n8n), LLM Prompt Engineering, Python, Email/API Integration

### Comprehensive Clinical Data Analysis & Visualization

2025

**Stack:** Python, Pandas, Matplotlib/Seaborn, Scikit-learn

- Performed end-to-end exploratory data analysis (EDA) on raw clinical datasets, including cleaning, transformation, and feature engineering.
- Built visualizations of clinical insights such as patient distribution by disease and other key trends across the dataset.
- Applied machine learning techniques to generate predictive insights from patient records.

**Skills Applied:** Exploratory Data Analysis, Feature Engineering, Pandas, Data Visualization (Matplotlib/Seaborn), Scikit-learn, Predictive Modeling

### AI PDF Question-Answering Platform

2025

**Stack:** Python, FastAPI, LLM APIs

- Built a tool that allows users to upload a PDF and ask natural-language questions, returning direct answers sourced from the document.

**Skills Applied:** FastAPI, Document Parsing, LLM API Integration, REST API Design

### Machine Learning Models Development

2024 – 2025

- Trained 5+ classification and regression models with a full preprocessing pipeline: encoding, normalization, scaling, and train/test splitting.
- Evaluated models using accuracy, F1-score, precision/recall, and confusion matrices; selected optimal algorithms per dataset characteristics.

**Skills Applied:** Supervised & Unsupervised Learning, Model Evaluation, Scikit-learn, Feature Engineering

### RESTful API Development & ML Integration

2024

- Engineered FastAPI-based REST APIs (GET/POST/PUT/DELETE) with request validation, integrating trained ML models as live prediction endpoints.

**Skills Applied:** FastAPI, REST API Design, ML Model Deployment, Request Validation

## CERTIFICATIONS

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- AI Fundamentals & Cloud — AWS
- AI Workflow: Feature Engineering & Bias Detection — IBM
- Building Systems with the ChatGPT API — DeepLearning.AI
- Machine Learning for All — University of London
- Machine Learning — Simplilearn
- App Deployment & Security — Packt
- Learn to Code with AI — Scrimba